

**ĐẠI HỌC QUỐC GIA THÀNH PHỐ HỒ CHÍ MINH**  
**TRƯỜNG ĐẠI HỌC CÔNG NGHỆ THÔNG TIN**  
**KHOA KHOA HỌC MÁY TÍNH**



*Khai thác dữ liệu và Ứng dụng*

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**BÁO CÁO**  
**Đồ án CS313.[XX]**

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**Giảng viên hướng dẫn:** [INSTRUCTOR NAME]

**Sinh viên thực hiện:** [STUDENT 1 — ID]  
[STUDENT 2 — ID]  
[STUDENT 3 — ID]  
[STUDENT 4 — ID]  
[STUDENT 5 — ID]

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# 1 Data Introduction

This project is built on **EdNet-KT4**, the largest publicly available educational interaction dataset, released by Riiid AI Research and collected from *Santa*, a Korean TOEIC preparation platform. KT4 is the most detailed of the four EdNet sub-datasets: it logs every fine-grained student action — not only question responses, but also page navigation, audio/video playback events, payments, and coupon redemptions. This level of detail makes it a strong fit for a CS313 data-mining project: the full pipeline (cleaning, integration, transformation, reduction, feature engineering, and modeling) can be exercised on a real, large-scale, multi-modal log.

## 1.1 Scale and provenance

| Property             | Value                                    |
|----------------------|------------------------------------------|
| Total interactions   | 131,441,538 rows                         |
| Total users          | 297,915 unique students                  |
| Collection period    | 2018-08-27 to 2019-12-01 (461 days)      |
| Source platform      | Santa (Riiid) — TOEIC AI tutor           |
| Raw size             | 6.4 GB across 297,915 per-user CSV files |
| Consolidated parquet | 1.3 GB (snappy-compressed, single file)  |

**Table 1:** EdNet-KT4 at a glance.

## 1.2 Schema

The interaction log has 8 columns. Four metadata files (questions, lectures, payments, coupons) provide the content side of the dataset.

| Column      | Type     | Description                                              |
|-------------|----------|----------------------------------------------------------|
| timestamp   | int64    | Unix ms; shifted from real values for privacy            |
| action_type | category | 13 types (enter, respond, play_audio, ...)               |
| item_id     | string   | Question / bundle / lecture / payment / coupon           |
| cursor_time | float64  | Media playback position in ms                            |
| source      | category | 8 in-app contexts (sprint, my_note, adaptive_offer, ...) |
| user_answer | category | a / b / c / d                                            |
| platform    | category | mobile / web                                             |
| user_id     | int32    | Anonymized student identifier                            |

**Table 2:** Schema of the consolidated interaction log.

The companion `questions.csv` (13,169 rows) records the question’s TOEIC part (1–7), correct answer, and skill tags; `lectures.csv` (1,021 rows) records the lecture’s part, tags, and video length. Together they enable the integration step that yields a correctness label for every respond action.

## 1.3 Why this dataset

EdNet-KT4 is appealing for our project for three reasons. First, its **raw scale** (131M rows, 6.4 GB) forces us to use efficient columnar formats and out-of-core operations from the very first step. Second, its **action diversity** — beyond just answers — gives the preprocessing and feature-engineering stages real material to work on (time gaps, fatigue proxies, lecture engagement, behavioral signals). Third, the dataset has a natural **prediction target**: given a student’s history, can we predict whether their next response will be correct? This is a well-studied knowledge-tracing problem on which we can compare classical (tree-based) and sequence (LSTM, 1D-CNN) models on the same evaluation protocol.

# 2 Exploratory Data Analysis

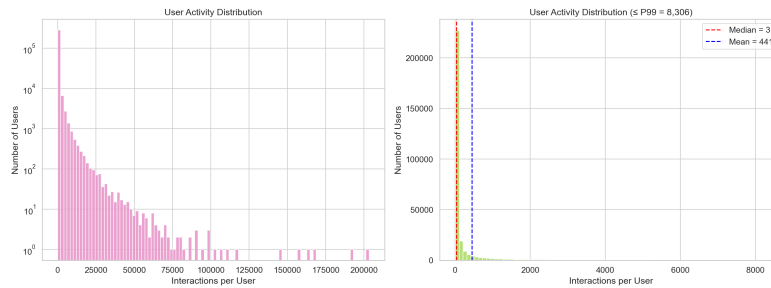
The EDA was structured into four passes, mirroring the natural questions a data miner asks when meeting a dataset for the first time: structural, distributional, content-metadata, and temporal. Full reports and 26 plots live in `output/eda/`; this section consolidates the most load-bearing findings.

## 2.1 Structural overview

Of the 8 columns, three carry the bulk of the missing-value mass — `cursor_time` (71.6% missing), `user_answer` (78.5%), and `source/platform` (0.02% each). Crucially, all of these are **structurally missing**: `cursor_time` is only emitted by media actions, `user_answer` only by respond/erase actions, and `source/platform` are null only on payment-related actions (no learning context). Within their applicable subsets they are 100% present. This means **no imputation is needed** — filling these would inject misleading information rather than recover lost signal.

## 2.2 Distributions and the long-tail of users

User activity is extremely right-skewed: the median user has only **31 interactions**, while the mean is **441**. The top 1% of users contribute 8,300+ interactions each, dragging the mean up by an order of magnitude. This has direct consequences for downstream modeling: cold-start users (P25 = 22 interactions) carry very little behavioral history, while a small power-user cohort dominates any naive aggregate statistic.



**Figure 1:** User-level interaction count distribution (log-y). The long tail makes cold-start vs. power-user a first-class modeling concern.

Other distributional findings worth flagging:

- **Sprint mode dominates** (71.3% of all actions). Students prefer self-directed part-specific practice; the AI’s `adaptive_offer` accounts for only 6.6%, and `tutor` just 0.9%. Most learning is student-driven, not platform-driven.
- **Mobile/web split: 71/29**, stable across nearly every action type. The single exception is `undo_erase_choice` — 142,092 events on web, **exactly zero on mobile** — indicating the feature was never shipped on the mobile UI.
- **Audio dominates over video** ( 25% of all actions vs. 3%), consistent with TOEIC’s listening-heavy format.

## 2.3 Content metadata: questions, parts, difficulty

The 13,169 questions are heavily concentrated in **Part 5 (Incomplete Sentences)**, which holds 43.3% of the bank — nearly half of all questions are grammar/vocabulary fill-in-the-blank. Listening parts 1–4 are smaller individually but use bundled passages (multiple questions per audio clip).

Difficulty was estimated as 1 — accuracy rate across all student responses. Across 23.3M matched responses, **overall accuracy is 56.87%**; difficulty is approximately normal around 0.39, with very few trivially easy or impossibly hard items.

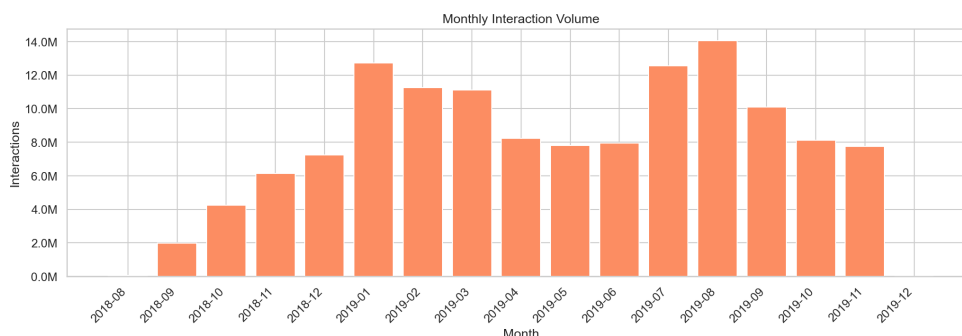
| Part | Section               | Avg. difficulty | Accuracy |
|------|-----------------------|-----------------|----------|
| 5    | Incomplete Sentences  | <b>0.487</b>    | 51.3%    |
| 6    | Text Completion       | 0.404           | 59.6%    |
| 7    | Reading Comprehension | 0.393           | 60.7%    |
| 4    | Short Talks           | 0.377           | 62.3%    |
| 2    | Question-Response     | 0.369           | 63.1%    |
| 3    | Short Conversations   | 0.364           | 63.7%    |
| 1    | Photo Descriptions    | <b>0.326</b>    | 67.4%    |

**Table 3:** Difficulty by TOEIC part. Part 5 is both the largest and the hardest section.

The combination is striking: Part 5 is simultaneously the *largest* share of the question bank (43%) and the *hardest* to answer (51.3% accuracy). Students spend the most time exactly where they perform the worst.

## 2.4 Temporal patterns

Daily activity exhibits a clear bimodal pattern across the 461-day collection window, with peaks in **January 2019** and **July–August 2019**. These align with common TOEIC exam preparation cycles in Korea — students ramp up usage in the months preceding major test sittings. August 2019 is the busiest single month at 14M interactions.



**Figure 2:** Monthly interaction volume. Bimodal peaks (Jan 2019, Jul–Aug 2019) align with TOEIC exam-prep seasons.

Day-of-week and hour-of-day analyses show **weekday domination** (Tuesday is the busiest day, Sunday the quietest, 23% lower) and a **single-peak daily cycle** — there is no secondary evening study peak. Daily Active Users averages just 2,013 against the 297K total user base ( 0.7% DAU/MAU), typical for a freemium edtech product with high churn outside exam season.

## 3 Preprocessing

The preprocessing pipeline is structured around the four classical data-mining stages from the CS313 curriculum: **cleaning**, **integration**, **transformation**, and **reduction**. The full implementation lives in `preprocessing/preprocessing.py`. The input is the consolidated 131M-row interaction parquet, the output is `kt4_preprocessed.parquet` (131.0M rows, 30 columns, 2.4 GB).

### 3.1 Cleaning

**Missing values.** As established during EDA, all four columns with missing values are structurally missing — present 100% of the time when applicable. No imputation is applied.

**Duplicates.** 461,237 exact duplicate rows (0.35%) were detected and removed. These almost certainly originate from client-side retry logic (network blips re-sending the same event). Removal is lossless.

**Consistency.** Programmatic checks confirmed that `user_answer` values are always in {a, b, c, d}, `platform` is always in {mobile, web} (or null where structurally expected), no timestamps are non-positive, and per-user event timestamps are monotonically non-decreasing in a 10K-user sample. **Zero violations** found.

**Outliers (cursor\_time).** Media-action playback positions range up to 11.6M ms ( 3.2 hours) — clearly accidental left-running sessions. Using the IQR rule on the non-null subset (Q1 = 0, Q3 = 17,650 ms, upper bound = 44,125 ms), **2.68M extreme cursor\_time values were set to NaN**. The rows themselves are preserved — only the unreliable measurement is cleared, so the subsequent min-max normalization sees a meaningful 0–44s range instead of being crushed by 3-hour outliers.

### 3.2 Integration

We derived `item_type` from the prefix of `item_id` (q/b/l/e/p/c) and used it to drive joins with the metadata files:

- **Questions join:** 28,142,959 question-related rows matched against `questions.csv` at **100%** success; added `bundle_id`, `correct_answer`, `part`, `tags`.
- **Lectures join:** 5,009,098 lecture-related rows matched against `lectures.csv` at **100%** success; added `lecture_part`, `lecture_tags`, `lecture_video_length`.

- **Correctness:** `is_correct = (user_answer == correct_answer)` computed for every response action — 23,308,702 labeled rows in total, with **56.87% correct**. This single column is the foundation for every downstream prediction task.
- **Redundancy:** `explanation_id` was found to always equal `bundle_id` for every question and was therefore not added; `bundle_id` and `item_type` were kept even though derivable, because they avoid repeated joins downstream.

### 3.3 Transformation

**Attribute construction:** from `timestamp` we derived `hour`, `day_of_week`, `date`, `time_since_prev` (per-user inter-event gap in ms), and `action_seq` (per-user 0-indexed action number).

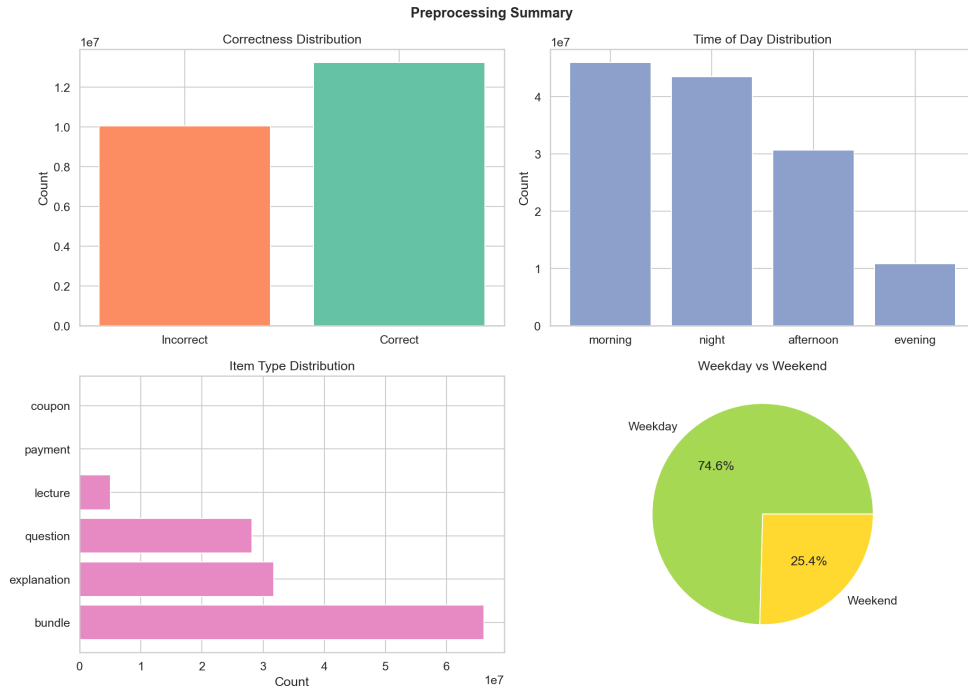
**Normalization:** `cursor_time_normalized` is `cursor_time` min-max-scaled to  $[0, 1]$  over the cleaned 0–44s range; `log_time_since_prev = log(1 + time_since_prev)` tames the extreme right skew (median 3 s, max 446 days) into a distribution amenable to downstream models.

**Discretization:** `time_of_day` bins `hour` into {night, morning, afternoon, evening}; `is_weekend` binary-encodes Sat/Sun.

**Encoding:** integer label encoding for the four categorical columns (`action_type`, `source`, `platform`, `item_type`); the original strings are preserved alongside.

### 3.4 Reduction (deferred)

The full preprocessed dataset is saved without reduction so the team retains flexibility for downstream tasks. The report documents six reduction strategies for downstream use — user sampling, stratified sampling, attribute subset selection, aggregation, cold-start filtering, and PCA — to be invoked task-by-task rather than baked into the canonical preprocessed file.



**Figure 3:** Preprocessing pipeline summary: input (131.4M, 8 cols) → output (131.0M, 30 cols).

## 4 Feature Engineering

The feature-engineering module transforms the preprocessed interaction log into a row-per-response feature table for the correctness-prediction task. Each row corresponds to one student response, enriched with **10 engineered features** grouped into four categories: question properties, student mastery, behavioral signals, and engagement/fatigue.

**Script:** `feature_engineering/generate_features.py`. **Output:** `kt4_features.parquet` — 23,308,702 rows × 15 columns ( $\approx 484$  MB).

## 4.1 Leakage prevention

All cumulative student features are computed using *only* data from interactions  $1, \dots, t-1$  when scoring time  $t$ . The first row of each user’s history has all cumulative features explicitly set to 0 (no prior history available). Validation confirms zero users with non-zero accuracy at their first response — i.e., the leakage-free property holds end-to-end.

## 4.2 The ten features

| Feature                  | Corr.  | Group      | Description                                        |
|--------------------------|--------|------------|----------------------------------------------------|
| feat_question_difficulty | −0.300 | Question   | 1 − accuracy for the question, across all students |
| feat_overall_accuracy    | +0.140 | Mastery    | Cumulative student accuracy through $t-1$          |
| feat_reading_accuracy    | +0.112 | Mastery    | Cumulative accuracy on Parts 5–7                   |
| feat_listening_accuracy  | +0.097 | Mastery    | Cumulative accuracy on Parts 1–4                   |
| feat_is_rapid_guess      | −0.052 | Behavioral | Response faster than P10 (690 ms): likely guess    |
| feat_log_session_fatigue | +0.038 | Engagement | log1p of action count in trailing 1-hour window    |
| feat_answer_changes      | −0.023 | Behavioral | Prior respond actions on this same question        |
| feat_adaptive_ratio      | +0.022 | Behavioral | Share of prior attempts coming from adaptive_offer |
| feat_lecture_watches     | +0.017 | Engagement | Cumulative lectures consumed before $t$            |
| feat_total_attempts      | −0.010 | Mastery    | Cumulative respond count through $t-1$             |

**Table 4:** Engineered features, ranked by absolute correlation with the target.

## 4.3 Group rationale

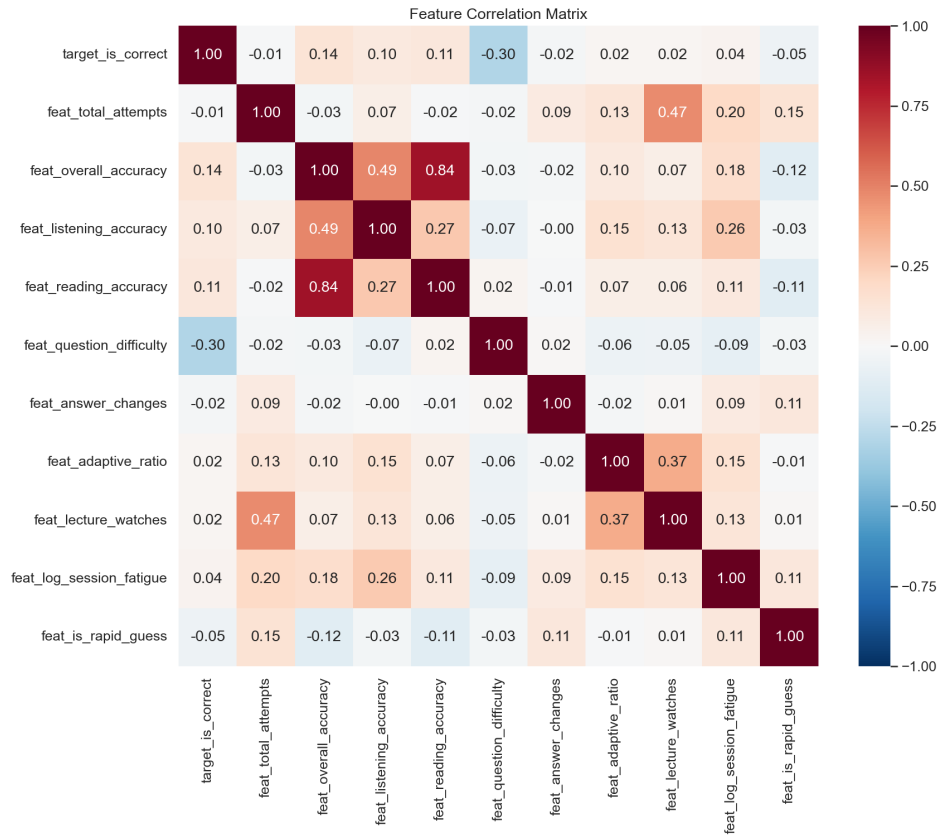
**Question properties.** `feat_question_difficulty` is by a wide margin the single most predictive signal ( $|\rho| = 0.30$ , roughly  $2\times$  the next strongest feature). It is an item-level property — independent of the current student’s history — which means no leakage and no temporal staleness.

**Student mastery.** The three accuracy features capture knowledge state at three granularities. The split between `feat_listening_accuracy` (Parts 1–4) and `feat_reading_accuracy` (Parts 5–7) has a mutual correlation of only **0.27**, confirming that TOEIC’s two halves test genuinely distinct competencies. `feat_total_attempts` provides a confidence-weighting signal — an accuracy of 0.5 from 2 attempts is very different from 0.5 from 500 attempts.

**Behavioral signals.** `feat_answer_changes` is unique to the KT4-flavored dataset (other EdNet variants don’t expose re-encounters at this granularity); it captures uncertainty and revision behavior. `feat_is_rapid_guess` flags responses under 690 ms (P10 of response time): rapid-guess responses are  $\sim 9$  percentage points less accurate. `feat_adaptive_ratio` quantifies the student’s reliance on platform recommendations vs. self-direction.

**Engagement & fatigue.** `feat_log_session_fatigue` counts *all* actions (not just responses) in the trailing 1-hour window before each response, log-transformed to handle the heavy right tail. `feat_lecture_watches` captures content consumption that may causally improve later accuracy.

## 4.4 Validation



**Figure 4:** Feature correlation matrix. Strongest negative correlation with the target is `feat_question_difficulty` ( $-0.30$ ); the listening/reading accuracy split is informative (mutual  $\rho = 0.27$ ).

Final integrity checks: zero null values across all 15 columns, `feat_total_attempts` monotonically increasing per user, `feat_lecture_watches` cumulative per user (not lifetime constant), and row count exactly matching the count of preprocessed respond actions (23,308,702). The feature table is now ready to feed the modeling stage.